

Akhilesh Kumar Ramachandran

+44 7466 612976 | akhil24.kumar@gmail.com | Google Scholar | LinkedIn

EDUCATION

Ph.D. in Sports Biomechanics Cardiff Metropolitan University , UK	2022 – 2026
M.Sc in Sports Biomechanics Loughborough University , UK	2017 – 2018
B.E in Electrical & Electronics Engineering Anna University , India	2011 – 2015

PROFESSIONAL DEVELOPMENT

- Performance Enhancement Specialization, **National Academy of Sports Medicine**, USA | 2017 – 2018
- Anatomy and Physiology, **Oregon Institute of Technology**, USA | 2017
- Exercise Physiology, **Massachusetts General Hospital**, USA | 2016
- Level 1 Strength and Conditioning, **Australian Strength and Conditioning Association**, Australia | 2015

PROFESSIONAL EXPERIENCE

Senior Research Assistant Mar' 2026 – Present

Cardiff Metropolitan University, UK

- Contributed to the design and delivery of a three-round Delphi consensus study on return-to-sport decision-making after ACL injury and reconstruction, informing development of more consistent and evidence-informed rehabilitation progression criteria.
- Supported the design of clinician interviews and co-design workshops to develop a clinically feasible ACL prehabilitation pathway for patients awaiting ACL reconstruction.
- Prepared stakeholder-facing outputs to translate ACL rehabilitation findings into practical recommendations for clinicians and service-development partners.

Research Associate Dec' 2025 – Current

Cardiff Metropolitan University, UK

- Led evaluation of the Twenty20 business growth and leadership programme, analysing turnover and stakeholder engagement data to evidence business growth and economic impact.
- Evaluated the impact of creative digital inclusion workshops across five Welsh schools on pupils' digital skills, confidence, and participation.

Doctoral Researcher Apr' 2022 – May' 2026

Cardiff Metropolitan University, UK

- Analysed longitudinal athlete datasets using generalised linear mixed-effects models to quantify interaction effects between physical load, biological maturation, and injury risk in youth female gymnasts.
- Designed and executed controlled studies with 100+ participants to identify biomechanical ACL injury risk patterns, applying mixed-methods ANOVA and translating findings into actionable screening strategies.
- Led human-centered biomechanical studies with 30+ diverse participants to validate Theia3D markerless motion-capture system, quantifying measurement precision and errors to inform system improvements and potential commercial applications.
- Developed and deployed Meta Quest 3 VR prototypes delivering real-time kinematic feedback during jump-landing tasks, measuring the effect of visual augmentation on lower-limb biomechanics.
- Built a custom Python signal-processing pipeline for cut-off frequency optimisation, improving measurement accuracy in high-frequency kinematic time-series data across 1,500+ movement trials.
- Automated quantitative analysis workflows reducing processing time by 70%, enabling faster delivery of evidence-based recommendations to applied sport and clinical teams.

Research Associate Jan' 2024 – Mar' 2025

Cardiff University, UK

- Led interviews with 10+ patients to explore experiences and expectations of NHS prehabilitation services, highlighting key themes in patient motivation and service accessibility.
- Generated actionable insights to improve service delivery and staff training through analysis of clinician-patient interactions across 40+ prehabilitation sessions.

- Conceptualised evidence-based toolkits to increase patient engagement with prehabilitation services, transforming research insights into accessible, patient-focused materials for clinicians.

Research Assistant

Jan' 2021 – Dec' 2022

University of Los Lagos, Chile

- Deployed wearable MEMS-GPS sensors across a 48-week elite football season, managing and processing 418 longitudinal athlete records spanning physical load, cardiovascular intensity, sprint velocity, and acceleration.
- Developed quantitative data-processing frameworks to clean, model, and analyse GPS and biometric time-series data, generating personalised workload models that improved performance decisions for elite athletes.
- Demonstrated that training workload and maturity status jointly explained 55% of performance variability in under-16 elite soccer players using regression modelling.

Research Assistant

Jan' 2020 – Dec' 2022

Polio Australia, Australia

- Analysed ten years of longitudinal polio hospitalisation data using STATA to map population health trends, producing evidence that informed national public health strategy in Australia.
- Led investigation of exercise interventions for poliomyelitis survivors, evaluating the effects of muscle strengthening and cardiovascular fitness activities on rehabilitation outcomes.

U-21 Sports Science Intern

Jan' 2023 – Jun' 2023

Cardiff City Football Club, UK

- Collected and analysed player GPS and physical performance data, translating testing results into individualised load management recommendations for coaching and medical staff.

Strength & Conditioning Coach

Jun' 2015 – Nov' 2020

Sports Dynamix Private Limited, India

- Developed and implemented load-management and recovery programs for professional athletes at Olympic and international levels, enhancing recovery efficiency and reducing injury risk.
- Conducted sport-specific performance testing to assess physical capacity, monitor progress, and inform targeted training interventions for optimal competitive performance.
- Led presentation of training data analytics to coaches and performance lead, providing actionable insights that guided evidence-based decisions on athlete workload and recovery optimisation.

SCHOLASTIC AWARDS

- Vice-Chancellor Scholarship, **Cardiff Metropolitan University**, UK | 2022
- International School Scholarship, **Loughborough University**, UK | 2017
- Best Final Year Project for developing a voice-controlled robot, **Anna University**, India | 2015
- Runner-up, i-QUEST Annual Research Exhibition, **Anna University**, Chennai | 2012

LEADERSHIP

Student Warden

Jan' 2024 – Present

Cardiff Metropolitan University, UK

- Managed welfare and safeguarding cases for 500+ residents, coordinating with wellbeing services and specialist providers during emergencies and maintaining compliant case records.
- Facilitated communication between students and residence staff to resolve accommodation-related concerns and enhance student living experience.

Doctoral Student Representative

Sep' 2023 – Sep' 2024

Cardiff Metropolitan University, UK

- Represented doctoral researchers to university leadership and organised cross-departmental conferences and workshops to disseminate research findings to academic and practitioner audiences.
- Organised conferences and workshops for doctoral students to share their research findings.

PUBLICATIONS

Ramachandran AK et al. Influence of Neuromuscular Training Interventions on Jump-Landing Biomechanics and Implications for ACL Injuries in Youth Females: A Systematic Review and Meta-analysis. *Sports Med.* 2025.

- Lathlean TJH, Quadros N, **Ramachandran AK**, Jackson MJ. Post-polio hospital admissions in Australia over a 10-year period. *J Public Health (Oxf)*. 2025.
- Lathlean TJH, **Ramachandran AK**, Sim S, Whittle IR. Clinical utility and reliability of surface electromyography in chronic low back pain. *J Clin Neurosci*. 2024.
- Moeskops S et al., **Ramachandran AK** et al. Effects of Neuromuscular Training on Muscle Architecture in Trained Young Female Gymnasts. *J Strength Cond Res*. 2024.
- Ramachandran AK** et al. Changes in Lower Limb Biomechanics Across Various Stages of Maturation and Implications for ACL Injury Risk in Female Athletes. *Sports Med*. 2024.
- Singh U, **Ramachandran AK** et al. Influence of task and environmental constraints on batting and bowling performance in cricket. *Int J Sports Sci Coach*. 2023.
- Oliver JL, **Ramachandran AK** et al. Effects of Strength, Plyometric and Combined Training in High-Level Male Youth Soccer Players. *Sports Med*. 2024.
- Sundaram V, **Ramachandran AK** et al. Sex-based differences in concussion incidence and injury mechanism in team bat/stick sports. *Int J Sports Sci Coach*. 2023.
- Dudagoitia EB et al., **Ramachandran AK** et al. Jump rope training for health and fitness in school-age participants. *Int J Kinesiol Sports Sci*. 2023.
- Singh U, **Ramachandran AK** et al. Jump rope training effects on health- and sport-related physical fitness in young participants. *J Sports Sci*. 2022.
- Ramachandran AK**, Singh U, Lathlean TJ. Strength and conditioning practices for cricket fast bowlers. *Int J Sports Sci Coach*. 2022.
- Nobari H, **Ramachandran AK** et al. Quantification of Pre-Season and In-Season Training Intensity in Asian Professional Soccer Players. *Healthcare (Basel)*. 2022.
- Nobari H, **Ramachandran AK**, Oliveira R. Influence of opponent level on professional soccer players' training and match performance. *Human Movement*. 2023.
- Doma K, **Ramachandran AK** et al. Paradoxical Effect of Creatine Monohydrate on Muscle Damage Markers. *Sports Med*. 2022.
- Lathlean T, **Ramachandran AK** et al. Clinical utility and reproducibility of surface electromyography in chronic low back pain: protocol. *BMJ Open*. 2022.
- Nobari H, **Ramachandran AK** et al. Relationship between accumulated workload and change of direction ability in elite young soccer players. *Sustainability*. 2022.
- Ramachandran AK** et al. Effects of Plyometric Jump Training on Balance Performance in Healthy Participants. *Front Physiol*. 2021.
- Singh U, **Ramachandran AK**, Baxter BA, Allen SJ. Force-velocity-power relationship with sprint performance. *Sports Biomech*. 2024.
- Ramachandran AK**, Goodman SPJ, Jackson MJ, Lathlean TJH. Effects of muscle strengthening and cardiovascular fitness activities for poliomyelitis survivors. *J Rehabil Med*. 2021.
- Doma K, Gahreman D, **Ramachandran AK** et al. Leaf extract supplementation, exercise-induced muscle damage, and muscular performance. *J Sports Sci*. 2021.
- Ramachandran AK** et al. Biomechanical and physical determinants of bowling speed in cricket. *Sports Biomech*. 2024.

CONFERENCE PRESENTATION

- Ramachandran AK, Pedley JS, Moeskops S, Oliver JL, Myer GD, Lloyd RS. Changes in Lower Limb Biomechanics Across Various Stages of Maturation and Implications for ACL Injury Risk in Female Athletes: a Systematic Review. Techniquet. Cardiff, United Kingdom. July' 2023.

JOURNAL REVIEWER

- Journal of Sports Sciences
- Sports Biomechanics
- Scientific Reports
- Journal of Orthopaedic Surgery and Research
- Sport Sciences for Health
- BMC Sports Science, Medicine and Rehabilitation